

Diagnosis of Autism Spectrum Disorder After Age 5 in Children Evaluated Longitudinally Since Infancy

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Objective: The diagnosis of autism spectrum disorder (ASD) has been found to be remarkably stable but few studies have followed children not initially diagnosed with ASD beyond 3 years of age to examine late or delayed diagnoses. The present study used a prospective familial-risk design to identify children who had undergone multiple comprehensive assessments in preschool and were determined to be negative for ASD only to meet criteria for ASD when tested in middle childhood.

Method: Data were pooled across 3 research teams studying later-born siblings of children with ASD. Fourteen children met inclusion criteria for the late-diagnosed group and were compared with a large sample of high- and low-risk siblings from the same sites who had ASD or typical development (TD) outcomes at 3 years of age.

Results: As a group, the late-diagnosed children scored between the TD and ASD groups on most measures administered at 3 years and differed significantly from the ASD group on most measures. However, there was significant heterogeneity among the late-diagnosed cases. Seven showed very little evidence of ASD in preschool, whereas 7 demonstrated subtle, subthreshold symptomatology.

Conclusion: Some children with ASD might present with a subtle phenotype early in life or show a prolonged time course of symptom development. This emphasizes the need for screening and surveillance schedules that extend past 36 months and continued evaluation of any child who presents with atypical early development and/or high-risk status. The findings also shed light on reasons why the mean age of ASD diagnosis remains older than 4 years.

Key words: autism spectrum disorder, diagnosis, diagnostic stability

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The mean age of autism spectrum disorder (ASD) diagnosis in the United States is currently 4.3 years and has not changed appreciably in recent decades.¹ This indicates that diagnosis after the preschool years is not uncommon. In large part this is due to delay in referrals for diagnostic evaluation. Much research has been done to identify factors related to age at diagnosis, with the finding that minority race/ethnicity, lower socioeconomic status and parental education levels, residence outside a metropolitan area, and diagnostic overshadowing are significant contributors to delayed referral.² Of great interest are children who have had early developmental evaluations performed but still have a delay in recognition of ASD.

Many studies have examined the stability of an ASD diagnosis,³ but children not initially diagnosed have rarely been followed; that is, “stability” is generally equated with the positive predictive value of an initial diagnosis. Few studies have reassessed children not initially regarded as having ASD to provide insight into the phenomenon of late

or missed diagnoses. One study examined 692 consecutive referrals to a neurodevelopmental evaluation center, identified all children ($n = 18$) with a preschool diagnosis of severe receptive language delay, and reassessed them at a mean age of 8 years.⁴ All 18 met criteria for ASD at the later evaluation. Another study reported on 23 children, initially evaluated as preschoolers at an autism diagnostic clinic, who were reassessed at a mean age of 11 years.⁵ Of the 9 children who had not received an ASD diagnosis in preschool, 4 did so in school age. In the 2 studies, the children subsequently diagnosed with ASD had been identified with early delays in language or cognitive function, which could have overshadowed social impairments or made those impairments appear secondary.

The largest investigation was done by Davidovitch *et al.*⁶ who reviewed medical records, linked to a population-based registry, of 221 children diagnosed with ASD after 6 years of age who had been seen for developmental evaluation before 6 years and found to be negative

for ASD. Their initial preschool diagnoses included language, motor, and cognitive delays. Fewer than half the children had features suggestive of ASD in their preschool medical records. Davidovitch *et al.* offered 4 potential explanations for the delayed identification of ASD: diagnostic overshadowing by other conditions in the initial assessment, increasing symptoms over time, missed symptoms in the preschool evaluations, and inaccurate diagnosis after 6 years of age.

Prospective studies of high-risk infants offer a research strategy that could be informative to understanding why some children are diagnosed years after an initial ASD-negative evaluation. In such studies, infants at risk for ASD due to family history are enrolled in the first year of life and assessed for ASD multiple times in infancy and preschool, regardless of the presence or severity of symptoms. No clinical referral is required, which, particularly for children with less severe impairments, often would not occur until school entry.² Brian *et al.*⁷ followed such a high-risk cohort ($N = 67$) to middle childhood (mean age 9 years) and found 6 children who were considered not to have ASD at 3 years but later met diagnostic criteria for ASD. Shephard *et al.*⁸ similarly identified 5 children, from a sample of 42 high-risk siblings, who had not received an ASD diagnosis at 3 years but were diagnosed at 7 years.

The present study combined the later-diagnosed cases from the 2 previously published samples^{7,8} with those from a third site studying high-risk infant siblings to differentiate among potential explanations for delayed diagnosis.⁶ In addition to the power afforded by a larger sample, a strength of this study is the prospective design, which provides natural controls for factors that can limit access to diagnostic assessments² and follows children whether clinical concerns were present or not. The population-based mean age of diagnosis of 4.3 years¹ is driven largely by children who did not receive an early referral, not by children who were referred early and not diagnosed.⁶ Thus, focusing on a sample that was assessed for ASD at multiple time points, without an initial clinical referral, could be particularly informative to the pressing question of why some children are not diagnosed until middle childhood.

METHOD

Participants

The present analyses were carried out using data from 3 research teams whose procedures, measures, and assessment schedules were similar enough to permit data pooling. Informed consent was obtained at each site before data

collection, as was institutional review board approval to pool and analyze de-identified data across sites.

Participants were later-born biological siblings of children with ASD (high-risk group) or typical development (TD; low-risk group). Inclusion in the high-risk group required an ASD diagnosis, documented by a clinical report and/or an appropriate screening measure (eg, Social Communication Questionnaire [SCQ]), in the affected older sibling and no identified neurologic or genetic condition that could account for an ASD diagnosis (eg, fragile X syndrome). Inclusion criteria for the low-risk group were no family history of ASD in first- or second-degree relatives and at least 1 older sibling with TD, verified by intake questionnaire or standardized instrument (eg, SCQ). Additional inclusion criteria for the groups were birth after 35 weeks of gestation, maximum enrollment age of 18 months, initial outcome assessment at 3 years, and availability of later diagnostic assessments between 5 and 9 years. The present study focused on all children ($n = 14$) from the participating groups who did not receive a diagnosis at 3 years but met ASD criteria at a follow-up school-age visit. Comparison was made to the remaining children composing the samples at each site (470 high-risk siblings, 262 low-risk siblings).

Measures

Autism Diagnostic Observation Schedule. The Autism Diagnostic Observation Schedule (ADOS)⁹ is a standardized protocol that measures symptoms of ASD and has high inter-rater reliability and construct validity. It provides an overall total score (social affect + restricted and repetitive behaviors) with empirically derived cutoffs for autism spectrum and autism. A calibrated severity score, allowing comparison across modules, ranges from 1 to 10.¹⁰ The ADOS was administered at preschool (Modules 1 and 2) and school-aged assessments (Module 3) by clinical examiners trained to research-reliability standards who were unaware of previous diagnostic decisions.

Mullen Scales of Early Learning. This is a standardized developmental test for children from birth to 68 months that provides *T* scores (mean 50, SD 10) for nonverbal cognitive, receptive, and expressive language and for gross and fine motor skills.¹¹ The Mullen scales have good internal consistency and test-retest reliability.

Vineland Adaptive Behavior Scales. This parent-report measure assesses social, communication, motor, and daily living skills.¹² It is normed for use with infants to adults and provides standard scores and age equivalents. The Vineland was administered at all ages.

Social Communication Questionnaire. This brief parent-report questionnaire¹³ assesses the same social, communication, and repetitive behavior symptoms of ASD as the longer parent interview, the Autism Diagnostic Interview-Revised (ADI-R). Initial studies¹⁴ used a cutoff score of 15 as indicative of ASD but later investigations¹⁴ established use of this instrument for children as young as 2 years using a lower cutoff of 11. The SCQ was completed by parents of 3-year-old and school-age children. It was not administered at 1 site so these missing SCQ scores were estimated by transforming ADI-R total scores using linear regression.

Parent Concerns. At 3 years, parents were asked whether they had concerns about their child's behavior or development.¹⁵ Responses were classified by coders trained to 80% reliability into 1 of 7 categories of concern (social, language, repetitive behavior, motor, medical, temperament/behavior, regulatory). Primary variables of interest were total number of concerns and number of ASD-related concerns (a sum of social, language, and repetitive behavior concerns).

Preschool Outcome Classification. At the 3-year visit, participants were classified into 1 of 3 outcome groups using an algorithm previously adopted for high-risk cohorts.¹⁶ The TD outcome group ($n = 462$) did not meet criteria for ASD, had no more than 1 Mullen subtest score 1.5 SD below the mean, had no Mullen subtest scores 2 SD below the mean, and had an ADOS score that was at least 3 points below the autism spectrum cutoff. The non-TD group ($n = 185$) was composed of children who did not meet *DSM-IV* criteria for ASD but had low Mullen scores ($n = 30$) and/or high ADOS scores (within 3 points of the autism spectrum cutoff, $n = 72$; at or above the autism spectrum cutoff, $n = 83$). Obtaining an ADOS score higher than the autism spectrum cutoff was not sufficient for a diagnosis. Classification in the ASD group ($n = 99$) required an ADOS score at or above the autism spectrum cutoff and meeting *DSM-IV* diagnostic criteria for ASD using all available data. Highly trained clinical researchers (eg, licensed health professionals or research personnel with multiple years of experience in ASD) conducted the evaluations and determined outcome classifications.

School-Age Procedures. Participants were contacted and invited for an additional visit at 5 to 9 years of age (mean 7.4 years, SD 1.4 years); more than 80% of the sample agreed to participate and previously published analyses indicated no differences between those who were and those who were not followed at school age.^{8,17} All were administered the ADOS Module 3, an IQ test (an age-appropriate Wechsler scale [$n = 12$] or the Differential Abilities Scale [$n = 2$]), the Vineland, and the SCQ or the ADI-R. Outcome classifications used *DSM-5* diagnostic criteria and were conducted by

experienced clinical researchers unaware of previous assessment results. Fourteen participants, who are the focus of the present study, first met criteria for ASD *after* 3 years of age and had the diagnosis confirmed at their latest school-age visit (see Table S1, available online, for scores at mid-childhood). All sources of material collected at school age were used in making diagnoses, which is considered best practice.¹⁸ Three of the 14 late-diagnosed participants fell just shy of the ADOS cutoff but met *DSM-5* criteria based on parent interview and other information collected (ie, parent and examiner concerns, Vineland scores, SCQ or ADI-R scores; Table S1, available online).

RESULTS

Sample Characteristics at 3 Years

Table 1 presents descriptive statistics and assessment data for the late-diagnosed group and the 3 comparison groups (ASD, TD, non-TD) when they were 3 years old. The late-diagnosed group had significantly lower ADOS severity scores than the ASD and non-TD groups but significantly higher ADOS severity scores than the TD group. The late-diagnosed group had significantly higher Mullen scores than the ASD group and was not significantly different from the TD and non-TD groups. On the SCQ and the Vineland, the late-diagnosed group had scores that did not differ from those of the non-TD group but were significantly different from those of the TD and ASD groups. Thus, in general, the late-diagnosed group fell at an intermediate level between the ASD and TD groups. The one exception was in the area of parent concerns, in which the late-diagnosed group had significantly higher total and ASD-related concerns than the TD and non-TD groups and did not differ from the ASD group. Although the late-diagnosed group contained a larger proportion of girls than the ASD group, this difference was not statistically significant.

Our analytic plan included several methods to examine potential heterogeneity within the 3-year assessment data of the late-diagnosed group. We hypothesized that group findings might obscure individual differences within the 14 cases and possible explanations for the later diagnosis.⁶

Algorithmic Outcome Classifications

The first method used to parse heterogeneity was the outcome classification based on Mullen and ADOS scores, described earlier.¹⁶ Of the 14 children in the late-diagnosed group, 9 had a TD outcome at 36 months, whereas 5 were classified as non-TD, 3 because of high (but subclinical) ADOS scores and 2 because their ADOS scores were higher than the cutoff but did not meet *DSM* criteria for ASD at 36 months.

TABLE 1 Sample Characteristics at 3-Year Assessment

	Late-Diagnosed (n = 14)	ASD (n = 99)	Non-TD (n = 180)	TD (n = 453)
Age at 3-y visit (mo)	37.28 (2.2)	38.18 (2.8)	38.10 (2.5)	37.61 (2.3)
Age at first visit (mo)	6.86 (2.7)	8.23 (3.7)	7.18 (3.6)	7.20 (3.8)
Boys, %	57.14	73.74	56.67	47.02
ADOS calibrated severity score	2.71 (1.9)	6.89 (1.7)***	4.06 (1.8)***	1.29 (0.5)***
Mullen Expressive Language Age Eq	42.29 (8.9)	30.74 (10.4)***	37.82 (9.0)*	41.06 (7.1)
Mullen Receptive Language Age Eq	39.71 (6.0)	29.59 (11.5)***	37.38 (8.9)	40.82 (7.4)
Mullen Fine Motor Age Eq	36.07 (4.9)	31.31 (7.7)*	36.38 (8.3)	39.07 (6.5)
Mullen Visual Reception Age Eq	43.29 (8.5)	33.83 (12.0)***	41.14 (10.0)	44.94 (7.6)
SCQ total score	7.29 (5.9)	13.25 (6.4)***	5.38 (4.6)	3.54 (2.7)***
Vineland ABC	94.70 (12.7)	79.18 (13.1)***	94.88 (13.2)	101.79 (13.1)*
Parent ASD concerns ^a	1.10 (1.45)	1.29 (1.2)	0.50 (0.9)*	0.24 (0.6)***
Parent total concerns ^a	2.90 (3.0)	2.27 (1.8)	1.04 (1.4)***	0.61 (1.0)***

Note: ABC = Adaptive Behavior Composite; ADOS = Autism Diagnostic Observation Schedule; Age Eq = age equivalent; ASD = autism spectrum disorder; SCQ = Social Communication Questionnaire; TD = typically developing.

^aSample sizes for this variable: TD = 302; non-TD = 144; ASD = 48; late-diagnosed = 10.

* $p < .05$; *** $p < .001$; all comparisons versus late-diagnosed group.

Latent Class Analysis

The second method undertaken to understand phenotypic heterogeneity within the 14 late-diagnosed cases was latent class analysis (LCA), which was conducted on the entire sample. The flexmix program in the R statistical package (R Foundation, Vienna, Austria) was used to conduct an LCA on 5 variables: ADOS severity scores, verbal and nonverbal mental age from the Mullen, SCQ total score, and Vineland Adaptive Behavior Composite. The ADOS and SCQ scores were anchored at 0 and modeled as Poisson distributions to accommodate the significant right skew of the data.

Results showed no simple set of distinct classes in the data. The LCA found that up to 8 classes were distinguishable when using Akaike or Bayesian information criterion values for best solution determination. We focused on the 3-, 4-, and 5-class solutions as the most interpretable from a clinical standpoint (Table S2, available online, presents the cross-tabulation of outcome groups by LCA class membership). Regardless of which LCA solution was used, the late-diagnosed cases were distributed across all classes, affirming heterogeneity within the group and failing to find a single pattern that explained the late diagnoses.

Table 2 presents descriptive statistics for the 4-class solution, as an example. Class 1 had scores in the average range for the Mullen and Vineland scores and ADOS and SCQ scores well below the ASD cutoff, so this class could be interpreted as the TD class. Class 2 was symptomatic across all measures and could be interpreted as the ASD class. Class 3 was in the average to high average range on developmental measures and low on ASD symptom

measures and was interpreted as a TD class of high function. Class 4 was somewhat affected on all measures, but below ASD cutoffs on the SCQ and ADOS, and was interpreted as a broader phenotype class.

Parent Report Versus Clinical Assessment: Residuals Analyses

One issue suggested by the LCA and by previous studies¹⁹ is that there could be a group of children with scores that are discrepant across the directly administered measures and the parent-report instruments (ie, in which parents might report symptoms that were not apparent during clinical assessment or vice versa). For example, in the 4-class solution presented in Table 3, the parent-report scores (SCQ, Vineland) for class 4 are relatively more average than the clinical assessment scores (ADOS, Mullen). One hypothesis is that the late-diagnosed cases were not identified at 3 years of age because evidence across measures was not consistent, decreasing clinician confidence in a diagnosis. To examine this possibility, we calculated residuals for each child when regressing ADOS severity scores on SCQ totals, where the residual was taken to be an indicator of the disagreement between parent report and clinical assessment. A similar analysis was done using Vineland and Mullen language domain scores. Table 3 presents means and SDs of residuals for each outcome group, where more highly positive numbers mean higher scores on the parent-report measures versus clinical assessment and smaller magnitude or negative scores indicate lower scores on parent-reported instruments than direct assessments.

TABLE 2 Characteristics of Latent Classes From the 4-Class Solution at 3-Year Outcome

	Class 1 (n = 274)	Class 2 (n = 87)	Class 3 (n = 229)	Class 4 (n = 156)
Age at preschool outcome (mo)	37.08 (1.6)	38.29 (2.9)	38.69 (2.9)	37.58 (2.3)
Age at first visit (mo)	7.20 (3.7)	8.16 (4.2)	6.71 (3.1)	7.82 (3.9)
Boys, %	52.92	71.43	38.43	63.13
ADOS calibrated severity score	1.37 (0.6)	5.55 (2.9)	1.90 (1.4)	4.46 (2.2)
Mullen Expressive Language Age Eq	37.80 (4.1)	32.15 (12.4)	46.93 (6.4)	33.60 (6.5)
Mullen Receptive Language Age Eq	37.16 (4.5)	31.89 (13.3)	47.02 (6.0)	32.76 (6.3)
Mullen Fine Motor Age Eq	36.22 (4.8)	32.47 (9.3)	43.86 (6.0)	32.78 (5.5)
Mullen Visual Reception Age Eq	42.48 (6.4)	35.40 (13.1)	49.89 (6.3)	36.42 (7.9)
SCQ total score	3.56 (2.1)	16.98 (5.0)	2.74 (2.0)	5.94 (2.9)
Vineland ABC	98.33 (8.9)	76.84 (13.5)	108.33 (10.5)	89.75 (10.9)

Note: ABC = Adaptive Behavior Composite; ADOS = Autism Diagnostic Observation Schedule; Age Eq = age equivalent; SCQ = Social Communication Questionnaire; TD = typically developing.

For the SCQ versus ADOS residual analysis, the late-diagnosed group was marginally greater than 0 ($t = 1.66$, $p = .10$) and significantly greater than the non-TD or TD group ($p < .05$). However, the late-diagnosed group was not different from the ASD group, indicating that parents in both groups reported relatively more symptoms on the SCQ than on the ADOS compared with the TD and non-TD groups. For the Vineland versus Mullen language scores, the late-diagnosed group was not significantly different from any of the other 3 groups. Thus, the analyses of residuals did not point to a consistent pattern of discrepancies between parent report and directly administered measures as the reason the late-diagnosed group was not identified at 3 years of age. Similarly, class 4 from the LCA, which had the most discrepancies between parent report and directly administered measures, did not contain a disproportionate number of late-diagnosed participants. Residuals were actually largest for the ASD group, in which the lack of alignment between parent report and direct assessment apparently did not influence the diagnostic outcome determinations.

TABLE 3 Residual Analysis by Outcome Group

	Late-Diagnosed	ASD	Non-TD	TD
SCQ vs. ADOS	1.81 (6.5)	2.98 (6.2)	-1.68 (5.1)*	-0.40 (2.7)*
Vineland vs. Mullen language	-1.71 (13.1)	-5.42 (9.7)	-0.85 (10.9)	1.87 (9.9)

Note: ADOS = Autism Diagnostic Observation Schedule; ASD = autism spectrum disorder; SCQ = Social Communication Questionnaire; TD = typically developing.

* $p < .05$; all comparisons versus late-diagnosed group.

Longitudinal Change Analyses

A final analysis of heterogeneity within the late-diagnosed group focused on changes in their longitudinal phenotype from 3 years to the school-age visit. In each analysis described earlier, a subgroup of the late-diagnosed cases demonstrated typical or mostly typical development at 36 months, despite meeting criteria for ASD in middle childhood. To examine change over time in autism symptoms, we constructed a harmonized variable that included all social, communication, and repetitive behavior items that are consistent across ADOS Modules 1 and 2 (used at the 3-year visit) and Module 3 (used at the school-age visit). This permitted an examination of change from preschool to middle childhood on items that could be scored across the full age range of the sample (ie, symptoms that show developmental continuity). The items making up the harmonized composite are presented in Table S3 (available online).

We compared scores on the harmonized composite at 3 years and at the school-age visit. Seven of the 14 participants demonstrated a pattern in which ADOS scores significantly increased from 3 years to school age. The other 7 participants had scores on the harmonized composite that were stable across time periods. Table 4 presents descriptive characteristics of these "increase" and "stable" subgroups at their 3-year assessment. The 2 subgroups had similar scores on the Mullen, Vineland, SCQ, and parent concerns measures (all effect sizes in the small range; Cohen $d < 0.50$). However, the subgroups did differ significantly on the ADOS harmonized composite and standard calibrated severity scores, with the stable subgroup demonstrating significantly higher scores than the increase subgroup at 3 years.

Next, we performed an ADOS item analysis to examine whether there were particular patterns that were shared, and others that were distinct, between the 2 longitudinal change subgroups of the late-diagnosed group. Means and standard

TABLE 4 Characteristics of Longitudinal Change Subgroups at 3-Year Assessment

	Late-Diagnosed Stable (n = 7)	Late-Diagnosed Increase (n = 7)
Age at 3-y assessment (mo)	37.08 (1.7)	37.47 (2.8)
Age at first visit (mo)	6.26 (3.2)	7.46 (2.3)
Boys, %	71.43	42.86
Mullen Expressive Language Age Eq	43.29 (9.0)	41.29 (9.5)
Mullen Receptive Language Age Eq	39.71 (6.5)	39.71 (5.9)
Mullen Fine Motor Age Eq	36.29 (6.0)	35.86 (4.0)
Mullen Visual Reception Age Eq	44.14 (9.1)	42.43 (8.5)
SCQ total score	7.03 (6.3)	7.55 (6.0)
Vineland ABC	95.83 (8.3)	93.57 (16.6)
Parent ASD concerns ^a	1.40 (1.7)	0.80 (1.3)
Parent total concerns ^a	3.60 (3.4)	2.20 (2.8)
TD outcome at 3-y assessment, n (%)	3 (42.9%)	6 (85.7%)
ADOS harmonized composite at 3 y old	4.86 (2.0)	1.86 (1.1)**
ADOS calibrated severity score at 3 y old	3.86 (2.0)	1.57 (0.8)*
ADOS harmonized composite at school age	5.43 (1.7)	6.71 (1.4)
ADOS calibrated severity score at school age	5.29 (2.5)	6.00 (1.4)

Note: ABC = Adaptive Behavior Composite; ADOS = Autism Diagnostic Observation Schedule; Age Eq = age equivalent; ASD = autism spectrum disorder; SCQ = Social Communication Questionnaire; TD = typically developing.

^aSample sizes for this variable: stable subgroup = 5, increase subgroup = 5.

* $p < .05$; ** $p < .01$.

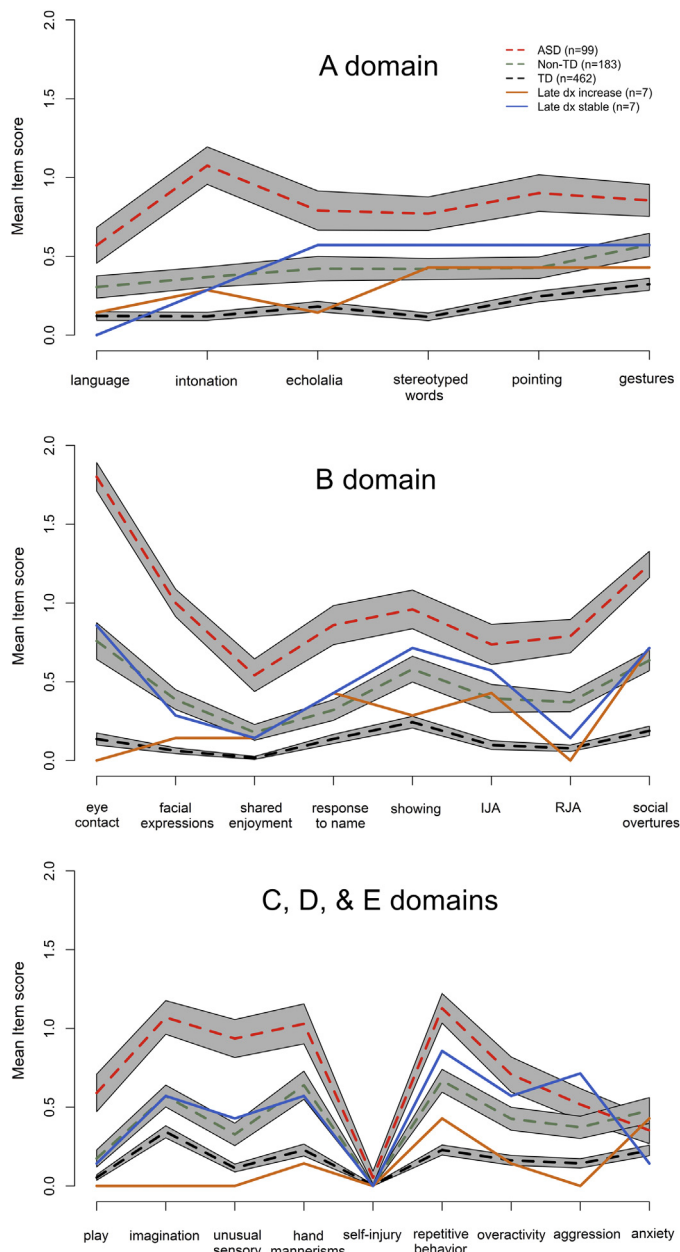
errors for all ADOS items at the 3-year assessment were calculated for the stable and increase subgroups. Figure 1 presents a comparison of the ASD, non-TD, and TD groups. The gray area around the ASD, non-TD, and TD group mean lines represents the 95% CI. As shown, neither the increase nor the stable subgroup was within the ASD CI on any ADOS item. The 2 subgroups had scores that were higher than those of the TD group on several ADOS items (stereotyped words/phrases, pointing, response to name, initiating joint attention, and quality of social overtures). There also was divergence between the subgroups, with the increase subgroup showing a typical pattern of scores at 3 years and the stable subgroup showing a more atypical pattern on several items (eg, echolalia, eye contact, showing, play, sensory, and hand mannerism items).

We also examined whether there were differences in ADOS item patterns between the increase and stable subgroups in middle childhood. We explored whether some children do not show typical early red flags (eg, deficits in joint attention, imitation, play) but only more subtle features that are hard to detect and might not be measured in preschool (eg, challenges with reciprocal conversation, empathy, circumscribed interests, etc.). Specifically, we examined whether the increase subgroup showed a disproportionate number of these later-appearing symptoms that are measured only on the higher modules of the ADOS (items: offers information, asks for information, reporting of events, empathy/comments on others' emotions, insight,

compulsions/rituals) to see whether these symptoms could account for the change and bring the participants up to diagnostic threshold. On the Module 3 items that are specific to later childhood and not scored in preschool, differences between the increase (n = 7, mean 5.1, SD 1.7) and stable (n = 7, mean 4.3, SD 1.6) subgroups were not statistically significant ($t_{12} = 0.98$, $p > .35$) and the effect size was moderate ($d = 0.48$), providing minimal support for this explanation.

DISCUSSION

This study sought to explain why children with known familial risk who had undergone multiple comprehensive assessments, with ASD as the primary outcome of interest, were determined to be negative for ASD in preschool, only to meet criteria when re-evaluated in middle childhood. Fourteen children from 3 international autism research teams were identified and compared with a large sample of high- and low-risk siblings from the same sites. The late-diagnosed group fell between the TD and ASD groups on most measures administered at 3 years and differed significantly from the ASD group on most variables. This indicates that there were real differences in their phenotypic presentation at 3 years from the early-diagnosed ASD cases and that the late-diagnosed group did not simply represent clinical errors (ie, "missed" cases). The one exception was on the parent concerns measure, in which the late-diagnosed group had scores that were significantly higher than those

FIGURE 1 Means for all Autism Diagnostic Observation Schedule Items at Age 3 by Group

Note: Gray areas around the ASD, non-TD, and TD group mean lines represent the 95% CI. ASD = autism spectrum disorder; IJA = initiating joint attention; RJA = response to joint attention; TD = typically developing.

of the TD and non-TD groups and statistically equivalent to those of the ASD group. This is consistent with a recent study reporting that parents of children with ASD might be able to detect clinically informative behaviors sooner than professionals¹⁹ and consistent with our finding that parents in the late-diagnosed and ASD groups reported more symptoms on the SCQ than were identified by standardized instruments such as the ADOS.

Davidovitch *et al.*⁶ described a group of children identified through a population-based registry who met criteria for ASD at school age after having been determined to not have ASD at previous evaluations. They proposed 4 explanations for such delayed diagnoses: diagnostic overshadowing from other conditions in the initial assessment, emerging or increasing symptoms over time, missed symptoms in the preschool evaluations, and inaccurate diagnosis at school age. Closer inspection of our data showed significant heterogeneity within the early presentations of the late-diagnosed participants. Using previously published algorithmic outcome definitions,¹⁶ 9 of the 14 late-diagnosed cases were classified as TD at 3 years. Using a latent class approach, 8 of the 14 were classified into TD classes. Using a longitudinal approach, 7 of the 14 were in a subgroup that showed significant increases, from typical to ASD levels, in their ADOS scores from preschool to middle childhood. Five of the 14 children were classified in the “typical” subgroup at 3 years by all 3 methods.

Davidovitch *et al.*'s first explanation for delayed identification, diagnostic overshadowing, suggests that other developmental challenges could have been more prominent and obscured detection of ASD. After examining our data, 5 of the 14 cases fell into non-typical outcome groups using the algorithm outcome definition.¹⁶ However, none of the 5 was formally diagnosed with another disorder. Exploration of clinical notes showed examiner concerns about language, social reticence or shyness, and behavioral rigidity, which are more suggestive of subthreshold symptoms than overshadowing by another disorder. Thus, although Davidovitch *et al.*'s diagnostic overshadowing explanation is consistent with the age of identification in the literature, in which the presence of other disorders has been shown to lead to later diagnosis,^{2,20} this pattern did not seem to fit any of our 14 cases.

Davidovitch *et al.*'s⁶ second potential explanation for delayed diagnosis was late symptom onset, in which “the child’s presentation changed or evolved, and ASD criteria were truly met at a later age after not having been present at an earlier age” (p. 232). This suggests that, in some children, the symptoms of ASD might not have fully emerged by 36 months. A priori, we expected few cases like this, because onset of ASD symptoms after 3 years has rarely been reported in the literature (other than children with childhood disintegrative disorder, who are generally much more affected and experience a catastrophic regression, unlike the present late-diagnosed cases). One way we examined this possibility was by exploring changes in behaviors key to diagnosis (eg, eye contact, gestures, directed vocalizations) that have developmental continuity and can be validly measured at all ages. Based on our longitudinal change analyses, 7 participants

demonstrated statistically significant increases in their ADOS scores over time from lower, non-ASD levels at 3 years, which could be taken as evidence of just such an emerging phenotype over time. This conclusion also was supported by the analysis of individual ADOS item patterns (Figure 1), which showed that the increase subgroup had scores outside the ASD range on all items at 3 years, even those measuring prototypic autism symptoms. Other studies have reported a possible continuum of onset timing, with symptoms appearing and unfolding at different rates across children.^{21,22} The marked changes in symptom presentation from preschool to school age in these 7 children imply that there is not a critical age window (ie, 3 years) before which ASD-related features are always manifest, suggesting that such symptoms can continue to evolve after this age. This is consistent with the *DSM-5*,²³ which no longer includes an age of onset criterion for diagnosis and acknowledges that symptoms might not be fully manifest until “social demands exceed limited capacity” (p. 50).

Davidovitch *et al.*'s⁶ third explanation is premised on underdiagnosis in preschool; that is, features of ASD were present but overlooked. This explanation is difficult to reconcile with the procedures used in the studies from which these data were drawn. All 14 children were seen at university autism centers, by clinical research teams with many years of expertise in early autism diagnosis, who performed multiple comprehensive ASD evaluations through 3 years of age. What is more likely, we believe, is that some children (those in the stable subgroup) demonstrated signs that were subthreshold at 3 years and did not evolve into impairment until environmental demands exceeded the child's abilities.²³ At 3 years, the child might not have been quite outside the range of typical development for social interest, communication, or behavioral rigidity, given the wide range of typical variation. However, by school age, if these mildly atypical early features had not resolved and information from new social contexts showed additional difficulties in social interaction, then the behaviors were judged by parents and clinicians as impairing and met diagnostic thresholds. This subgroup of 7 children might have had a more subtle presentation early in life, but some of the concerning behaviors did seem to have been present; then they intensified and came into sharper focus as more data from school and peer contexts accumulated. There also was a larger (but not statistically significant) proportion of girls in the late-diagnosed group and most participants in this group had Mullen scores in the average range at 3 years. If examiners had the expectancy that girls or cognitively average children were less likely to have ASD, then this also might have contributed to decreased clinician confidence at 3 years of age in some of these cases.

Davidovitch *et al.*'s fourth explanation is that ASD is overdiagnosed at later evaluations when parents seek out

and push for a diagnosis to improve service provision. This possibility cannot be completely refuted with the present study design, but we believe it is more likely to apply to a clinically referred sample⁶ than to a prospective sample. They also noted a “lack of reliance on formal assessments”⁶ for the later visits in which the diagnosis was made, which was not applicable to the present study.

The present study explored 2 other explanations for delayed diagnosis. First was the possibility that measurement discrepancies (ie, parent and examiner judgment) could lead to an initial reluctance to diagnose a symptomatic child and delay formal identification, but our data did not support this hypothesis. Second, we examined whether some children did not show typical early symptoms but only later-appearing features (eg, deficits in reciprocal conversation, unusual interests, etc.) that are not measured on the ADOS in the modules used with preschoolers.²⁴ Perhaps some children have intact skills in the areas that the early modules of the ADOS focus on (eg, joint attention, imitation, pretense) and deficits cannot be detected until instruments specifically designed to elicit the later-appearing signs are used. Our data did not provide strong support for this explanation either.

A major limitation of this study is the small size of the late-diagnosed group; therefore, our findings must be interpreted with caution and require replication in larger samples. We cannot rule out retention biases, in which parents are more likely to return for a school-age visit because of lingering worries about their child. For this reason, it is not possible to estimate the prevalence of late diagnosis in high-risk children from the present data. Despite the small sample, we conducted a number of post hoc tests to explore the behavioral phenotypes of the late-diagnosed children, given the unique nature of this group and its potential ability to shed light on factors affecting age of identification. The present findings highlight the challenges of evaluating young at-risk children who might present with a subtle phenotype early in life or show a prolonged time course of symptom development. Given the high-risk familial design of the present study, we do not know whether the results could generalize to low-risk groups, but the findings of Davidovitch *et al.*,⁶ which inspired our work, suggest that subthreshold presentations and late-emerging symptoms also are seen in the general population. This emphasizes the need for screening and surveillance schedules that extend past 36 months and continued evaluation of any child who presents with atypical early development and/or high-risk status. Indeed, with age at ASD diagnosis remaining stubbornly at older than 4 years internationally,^{1,2,25} despite apparent progress in discerning early signs, understanding the early development of children diagnosed in the school-age years remains an important priority.

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TABLE S1 Characteristics of the 14 Late-Diagnosed Children**In Middle Childhood**

ID	Sex	Risk Group	Age Diagnosed (mo)	Age at Latest Visit (mo)	Verbal IQ	Nonverbal IQ	ADOS Overall Total	ADOS Calibrated Severity Score	SCQ Total	Vineland ABC	Longitudinal Change Subgroup
Child 1	M	high	69	132	126	100	5	3	19 ^a	81	stable
Child 2	F	high	63	116	83	120	10	6	17 ^a	87	stable
Child 3	M	high	110	110	152	118	5	3	5 ^a	114	stable
Child 4	F	high	114	114	missing	missing	11	7	19 ^a	61	increase
Child 5	F	high	62	101	87	97	8	5	27 ^a	71	increase
Child 6	M	high	101	101	112	142	8	5	17 ^a	87	stable
Child 7	F	high	85	85	125	92	13	8	35	75	increase
Child 8	M	high	85	85	102	104	15	9	6	86	stable
Child 9	F	high	84	84	104	140	13	8	2	111	stable
Child 10	M	high	85	85	115	96	6	3	12	77	stable
Child 11	F	high	91	91	110	116	7	4	23	93	increase
Child 12	M	high	117	117	108	112	10	6	8	missing	increase
Child 13	M	low	84	84	120	105	11	7	8	90	increase
Child 14	M	high	94	94	105	117	8	5	5	missing	increase

At 3 y of Age

ID	Algorithmic Outcome Classification	ASD-Specific Parent Concerns, n	ADOS Calibrated Severity Score	ADOS Harmonized Score
Child 1	TD	0	2	2
Child 2	Non-TD	missing	5	8
Child 3	Non-TD	missing	4	6
Child 4	TD	missing	1	1
Child 5	TD	missing	1	2
Child 6	TD	4	2	5
Child 7	Non-TD	3	3	1
Child 8	Non-TD	1	7	6
Child 9	Non-TD	2	5	3
Child 10	TD	0	2	4
Child 11	TD	0	2	2
Child 12	TD	0	1	4
Child 13	TD	0	1	1
Child 14	TD	1	2	2

Note: ABC = Adaptive Behavior Composite; ADOS = Autism Diagnostic Observation Schedule; ASD = autism spectrum disorder; F = female; M = male; SCQ = Social Communication Questionnaire; TD = typically developing.

^aEstimated by linear regression from Autism Diagnostic Interview-Revised total scores.

TABLE S2 Latent Class Analysis Class Memberships by 36-Month Outcome

	Late-Diagnosed	ASD	Non-TD	TD
3-Class solution				
Class 1	4	29	108	66
Class 2	7	1	57	380
Class 3	3	69	15	7
4-Class solution				
Class 1	4	0	21	249
Class 2	3	63	15	6
Class 3	4	1	51	173
Class 4	3	35	93	25
5-Class solution				
Class 1	3	0	4	204
Class 2	2	9	81	4
Class 3	2	24	58	24
Class 4	4	0	23	217
Class 5	3	66	14	4

Note: ASD = autism spectrum disorder; TD = typically developing.

TABLE S3 Autism Diagnostic Observation Schedule Items Used in Harmonized Composite

	Module 1	Module 2	Module 3
Language and communication			
Overall level of non-echoed language	A1	A1	A1
Intonation of vocalizations/verbalizations	A3	A3	A2
Immediate echolalia	A4	A4	A3
Stereotyped/idiosyncratic words/phrases	A5	A5	A4
Gestures	A8	A8	A9
Reciprocal social interaction			
Unusual eye contact	B1	B1	B1
Facial expressions directed to others	B3	B2	B2
Shared enjoyment in interaction	B5	B3	B4
Quality of social overtures	B12	B8	B7
Play			
Imagination/creativity	C2	C2	C1
Stereotyped behaviors and restricted interests			
Unusual sensory interest	D1	D1	D1
Hand/finger/other complex mannerisms	D2	D2	D2
Self-injurious behavior	D3	D3	D3
Repetitive interests/stereotyped behavior	D4	D4	D4